



Operations Research

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Endterm - Exam SS25

Exam duration: 120 Minuten

Name:

Matriculation number:

Exercise 1 (16 Points). *Multiple-Choice Questions*

Mark all correct answers (there can be more than one):

Scoring: Each question is worth up to 2 points. You receive 2 points if you select all correct options and no incorrect ones. Selecting one incorrect option or not selecting a correct option reduces the score to 1 point. Any greater deviation or selecting no option results in 0 points.

(a) Which of the following statements are correct?

- ☐ Every linear program has a dual program.
- ☐ If a primal constraint is not tight, then its corresponding dual variable is zero.
- ☐ If the primal problem is infeasible, then the dual problem is unbounded.
- ☐ If the primal problem is unbounded, then the dual problem is infeasible.

(b) Which of the following statements about Christofides' algorithm (as presented in the lecture) are correct?

- ☐ The algorithm computes an Eulerian tour.
- ☐ The algorithm computes a Hamiltonian cycle.
- ☐ The algorithm applies only to metric TSP instances.
- ☐ The solution produced by the algorithm is at most twice as costly as any optimal solution.

(c) Which of the following statements about Cutting Planes are correct?

- ☐ Cutting planes remove integer solutions from the feasible region of the LP relaxation.
- ☐ An ideal outcome of cutting planes is the convex hull of all integer feasible solutions.
- ☐ The cutting plane method always ends with two different Gomory cuts.
- ☐ Combining cutting planes with Branch-and-Bound can speed up solving integer programs.

(d) Suppose you've solved an LP in standard form and found an optimal basis. Now you perturb the right-hand side of a constraint by Δ . Which of the following sensitivity statements are always true (assuming the basis stays optimal)?

- ☐ The shadow price vector π should be updated.
- ☐ The shadow price vector π can be used to find the change in the values of the basic variables.
- ☐ The shadow price vector π can be used to find the change in the value of the objective function.
- ☐ The solution can become degenerate for some Δ values.

(e) Let E be a finite set. For which of the following collections $I \subseteq \{S | S \subseteq E\}$ is (E, I) a matroid?

- ☐ $I = \{S \subseteq E : |S| \text{ is even}\}.$
- ☐ $I = \{S | S \subseteq E\}.$

- ☐ Let the ground set E be the edge set of an undirected graph $G = (V, E)$. Define $I = \{S \subseteq E : S \text{ contains no cycle in } G\}$.
- ☐ $I = \{S \subseteq E : |S| \text{ is odd}\}$.
- (f) Let $A \in \mathbb{R}^{m \times n}$ be a totally unimodular matrix and $b \in \mathbb{Z}^m$. Investigate the linear program $\max\{c^T x : Ax \leq b\}$. Which claims are correct?
- ☐ If c is integral, every optimal vertex solution x^* has integral value $c^T x^*$.
- ☐ Adapting A and b such that the constraint $x \geq 0$ is added can destroy total unimodularity of A .
- ☐ There exists a bipartite graph G such that A is the incidence matrix of G .
- ☐ The determinant of every submatrix of A has values in $\{+1, -1, 0\}$.
- (g) Which of the following scenarios can cause the simplex algorithm to cycle?
- ☐ The simplex algorithm never cycles.
- ☐ The feasible region is unbounded.
- ☐ There are multiple distinct optimal solutions ($x_i^* \neq x_j^*$)
- ☐ The problem is degenerate.
- (h) Which of the following are true?
- ☐ Gradient Descent methods always converge to the global minimum for strictly convex functions, independent of the chosen step-sizes.
- ☐ Gradient methods with momentum always need less iterations to converge than gradient methods without momentum.
- ☐ For a constrained optimization problem, a KKT point is always a local optimum.
- ☐ For a constrained convex optimization problem a KKT point is always a global minimizer, if Slater's condition holds.

Exercise 2 (16 Points). *Lemonade or Iced Tea*

A café produces two free drinks: *Lemonade* (x_1) and *Iced Tea* (x_2). Each batch of:

- Lemonade uses 4 L of water, 1 kg of sugar, and 1 kg of fresh lemon;
- Iced Tea uses 2 L of water, 2 kg of sugar, and 0.5 kg of tea leaves.

Available per day: 100 L water, 40 kg sugar, 30 kg lemons, 20 kg tea leaves.

The café wants to find a daily production plan that produces both drinks, so they solve the following feasibility problem:

$$\begin{aligned}
 &\max 0x_1 + 0x_2 \\
 &\text{s.t. } 4x_1 + 2x_2 \leq 100 && \text{(water)} \\
 &\quad 1x_1 + 2x_2 \leq 40 && \text{(sugar)} \\
 &\quad 1x_1 \leq 30 && \text{(lemons)} \\
 &\quad 0.5x_2 \leq 20 && \text{(tea leaves)} \\
 &\quad x_1, x_2 \geq 0.
 \end{aligned}$$

An employee introduces slack variables s_1, s_2, s_3, s_4 and arrives at the following tableau with the basis $\{x_1, x_2, s_3, s_4\}$, which is feasible but has *wrong Row 0 values*, since he has not taken an OR course in his life.

Basis	x_1	x_2	s_1	s_2	s_3	s_4	RHS
Row 0	0	-1	2	1	0	0	0
x_1	1	0	1/3	-1/3	0	0	20
x_2	0	1	-1/6	2/3	0	0	10
s_3	0	0	-1/3	1/3	1	0	10
s_4	0	0	1/12	-1/3	0	1	15

- What is the correct Row 0?
- After the correction of Row 0, is the tableau optimal? Why?
- The café wants to keep the price of Iced Tea at \$0 and find a price $\$p_1$ for Lemonade. Derive the range of p_1 for which the current basis remains optimal. Is it possible to find an optimal solution that sell Lemonade ($p_1 > 0$) and keep offering Iced Tea for free?
- Keeping all prices at zero, under what perturbation of the water capacity does the current BFS become *degenerate*? Show which basic variable first hits zero.
- Keeping all prices at zero, introduce a new drink, *Soft Churchill* that uses 3 L water and 1 kg lemon and would sell at $\$p_S$. Write its reduced cost formula using the optimal tableau from part b and state the condition on p_S for the new drink to enter the basis.

Exercise 3 (15 Points). *Duality*

Consider the following primal linear program (P):

$$\begin{aligned}
 &\text{Maximize} && 3x_1 + 2x_2 + x_3 \\
 &\text{subject to} && x_1 + 2x_2 + x_3 \leq 10 \\
 & && 2x_1 - x_2 + 2x_3 \geq 4 \\
 & && x_1 + x_2 - x_3 = 3 \\
 & && x_1, x_2, x_3 \geq 0.
 \end{aligned}$$

- Derive the dual linear program (D) corresponding to (P).
- State the primal and the dual complementary slackness conditions.
- Show that the primal solution $x = (2, 2, 1)^T$ is not optimal.

Exercise 4 (21 Points). *School Planning*

Welcome to *Newville* – a city so new it still smells like fresh paint. The city planners are deciding which schools to build. There are $I = \{1, \dots, 15\}$ possible school sites, each with a capacity of $c_i \in \mathbb{R}_{>0}$ and a construction cost of $f_i \in \mathbb{R}_{>0}$ Euros. The city has $J = \{1, \dots, 800\}$ students who need to be assigned to schools. Each student $j \in J$ has a preference score $s_{i,j} \in \mathbb{R}_{>0}$ for each school site $i \in I$, where $s_{i,j} > s_{i',j}$ indicates that student

j prefers school site i over i' . Your task is to help the city planners decide which schools to build and how to assign students to schools.

Let $y_i \in \{0, 1\}$ be a variable which is 1 if school $i \in I$ is built and 0 otherwise. Also, let $x_{i,j} \in \{0, 1\}$ be a variable which is 1 if student $j \in J$ is assigned to school $i \in I$ and 0 otherwise.

Formulate the following constraints as linear (in-)equalities. You may introduce additional variables. If you do so, please also state their intuitive meaning in words:

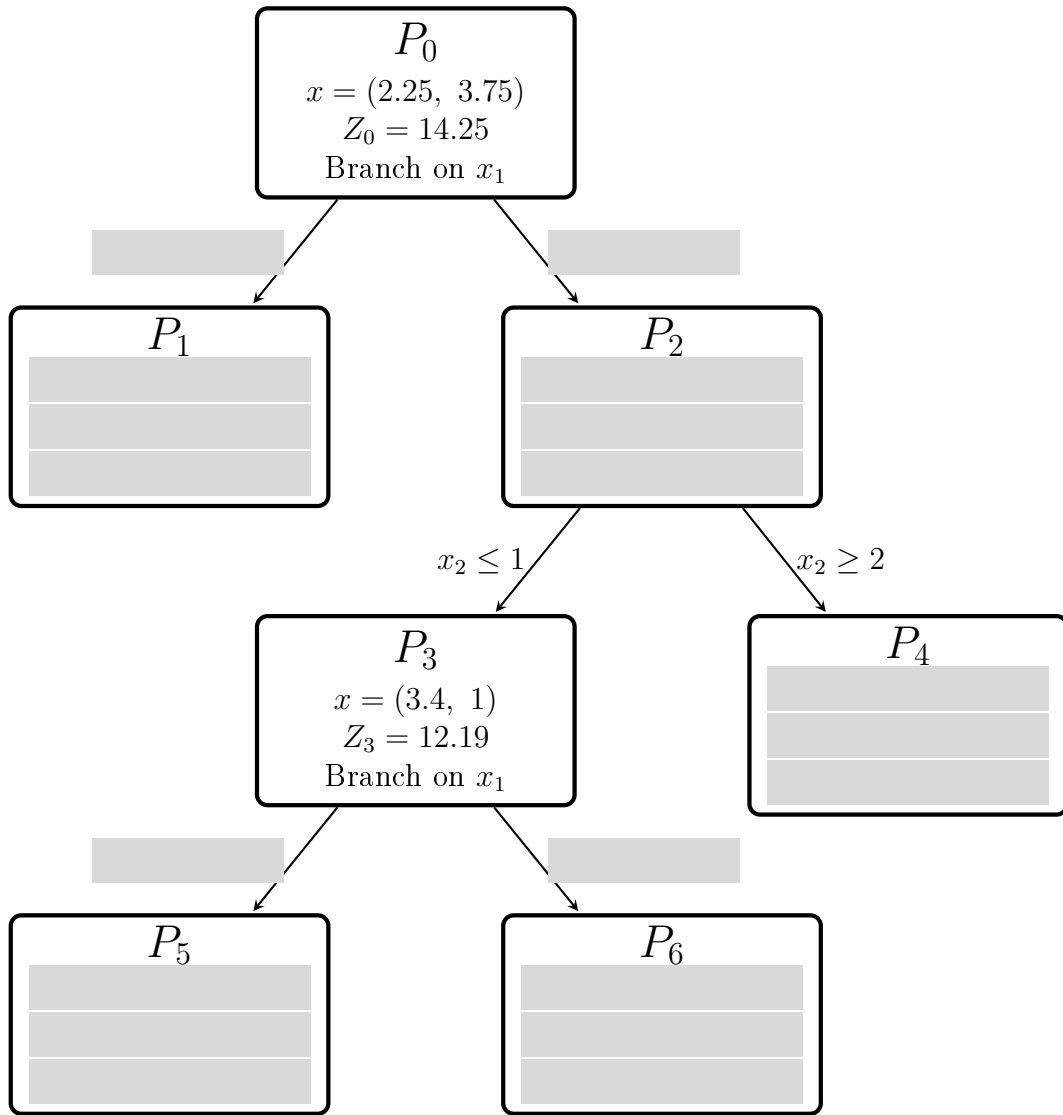
- a) Each student must be assigned to exactly one school.
- b) The total number of students assigned to each school cannot exceed the school's capacity.
- c) A student may be assigned to a school only if the school is built.
- d) Let the parameter $d_{i,i'} \in \mathbb{R}_{>0}$ denote the distance (in kilometers) between two school sites $i, i' \in I$ with $i \neq i'$. The distance between any two built schools must be at least 5 km.
- e) Assume that the city is divided into two zones, A and B . The parameter $A_i \in \{0, 1\}$ indicates whether school site $i \in I$ is located in zone A ($A_i = 1$) or in zone B ($A_i = 0$). If more than 200 students are assigned to schools in zone A , then at least 300 students must be assigned to schools in zone B .
- f) Formulate a single objective function that minimizes total construction costs while maximizing total preference scores of students.

Exercise 5 (14 Points). *Branch-and-Bound*

The great wizard *Gondolf* used the Branch-and-Bound method to solve the following integer program:

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 \\ \text{s.t.} \quad & 2.4x_1 + x_2 \leq 9.15 \\ & x_1 + x_2 \leq 6 \\ & x_1, x_2 \in \mathbb{N}_0. \end{aligned}$$

To document his journey, Gondolf sketched the branch tree he followed and compiled a table listing all LP relaxations he considered – including some he didn't end up using. In his notation, the subscript in a node name (e.g. P_1) refers to the constraint that was added, and whenever a branching choice was ambiguous, he chose to branch on the first variable. Unfortunately, Gondolf's clumsy friend *Balbo Buggins* spilled ink on the scroll, obscuring parts of the solution. Can you help recover the missing information and answer the following questions?



Subproblem	Constraints added to original LP	x_1	x_2	Z
P_A	-	2.25	3.75	14.25
P_B	$x_1 \leq 2$	2.00	4.00	14.00
P_C	$x_1 \geq 3$	3.00	1.95	12.90
P_D	$x_2 \leq 3$	2.56	3.00	13.69
P_E	$x_2 \geq 4$	2.00	4.00	14.00
P_F	$x_1 \leq 1, x_2 \geq 4$	1.00	5.00	13.00
P_G	$x_1 \leq 2, x_2 \geq 4$	2.00	4.00	14.00
P_H	$x_1 = 3, x_2 \leq 1$	3.00	1.00	11.00
P_I	$x_1 \geq 2, x_2 \leq 3$	2.56	3.00	13.69
P_J	$x_1 \geq 3, x_2 \leq 1$	3.40	1.00	12.19
P_K	$x_1 \geq 3, x_2 \leq 3$	3.00	1.95	12.90
P_L	$x_1 \geq 3, x_2 \geq 2$	Infeasible		
P_M	$x_1 \geq 4, x_2 \leq 1$	Infeasible		
P_N	$x_1 \geq 4, x_2 \leq 2$	Infeasible		
P_O	$x_1 \geq 4, x_2 \geq 2$	Infeasible		

- a) Identify which subproblems from the table correspond to the illegible nodes in the tree P_1, P_2, P_4, P_5, P_6 (e.g., $P_0 = P_A$).

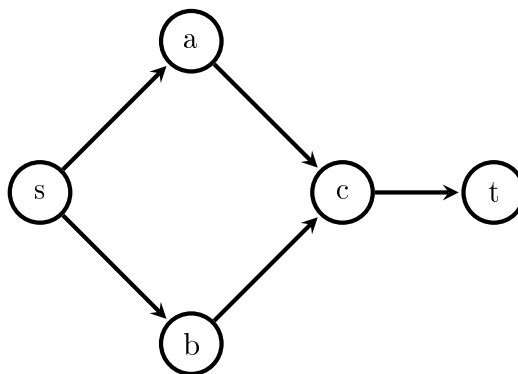
- b) Determine the missing constraints $P_0 \rightarrow P_1, P_0 \rightarrow P_2, P_3 \rightarrow P_5$, and $P_3 \rightarrow P_6$ (e.g. $x_2 \leq 1$).
- c) What are the values of x_1 and x_2 in the optimal integer solution, and what is the corresponding objective value?
- d) In which order would Gondolf have explored the nodes, assuming he used the following strategies (and stopped as soon as the optimal solution was confirmed) (e.g. P_0, P_2, P_3 , STOP)?
- (i) FIFO (First-In-First-Out)
 - (ii) LIFO (Last-In-First-Out)

Exercise 6 (14 Points). *Maximum flow and minimum cut counterexamples*

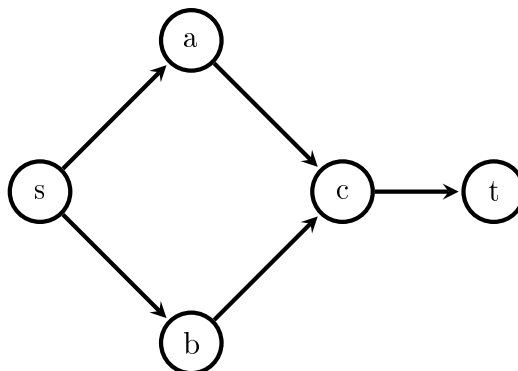
Let $(G = (V, E), s, t, c)$ be a network with source s , sink t , and integral capacities $c(e) \in \mathbb{N}$ for all edges $e \in E$.

Consider the following statements. Use the given graphs to find a suitable counterexample for each one of them. Clearly describe why the given counterexample refutes the statement.

- a) If all capacities are odd then there is a maximal $(s - t)$ -flow f such that $f(e)$ is odd for all $e \in E$.



- b) Adding a number $\lambda \in \mathbb{N}$ to all capacities $c(e)$ does not change the minimal cuts.



Exercise 7 (20 Points). *Minimal Circle Enclosure*

Matilda has been hired by Amazon to determine the optimal location for a new warehouse that will serve n distribution centers. Let the positions of the distribution centers be given by points $p_i = (x_i, y_i) \in \mathbb{R}^2$ for $i = 1, \dots, n$. Assume that at least two distribution centers have distinct locations, i.e., there exist j, k such that $p_j \neq p_k$.

Help Matilda find the optimal warehouse location $M = (c_1, c_2) \in \mathbb{R}^2$ that minimizes the maximum distance to any distribution center. In other words, find the center M and radius $r \geq 0$ of the smallest circle such that all points p_i lie inside or on the boundary of this circle.

- (a) State the definition of a convex function. Then, prove that any norm $\|\cdot\|$ on $\mathbb{R}^d$ is a convex function.

Hint: A norm satisfies the following properties for all $x, y \in \mathbb{R}^d$, and $\alpha \in \mathbb{R}$:

- Positive definiteness: $\|x\| \geq 0$, and $\|x\| = 0 \iff x = 0$.
- Homogeneity: $\|\alpha x\| = |\alpha| \cdot \|x\|$.
- Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$.

- (b) Formulate the problem of finding the smallest enclosing circle as a convex optimization problem. Specifically, express it in the form:

$$\begin{aligned} \min_{r, c_1, c_2} \quad & f(r, c_1, c_2) \\ \text{s.t.} \quad & g_i(r, c_1, c_2) \leq 0 \quad \text{for all } i = 1, \dots, n \\ & r \geq 0, \quad c_1, c_2 \in \mathbb{R} \end{aligned}$$

Show that the objective function f and the constraint functions g_i are convex.

Hint: Use the Euclidean norm $\|\cdot\|_2$.

- (c) Show that Slater's condition holds for this problem. That is, find a specific point (r, c_1, c_2) , expressed in terms of the given $\{(x_i, y_i)\}$, such that all inequality constraints are satisfied strictly.
- (d) Given the Lagrangian function:

$$L = f + \sum_{i=1}^n \lambda_i g_i - \mu r, \quad \text{with } \lambda_i, \mu \geq 0$$

use the **(KKT) conditions** to show that at the optimal solution (c_1^*, c_2^*) , the following must hold:

$$\begin{aligned} 1 &= \sum_{i=1}^n \lambda_i \\ c_1^* &= \sum_{i=1}^n \lambda_i x_i \\ c_2^* &= \sum_{i=1}^n \lambda_i y_i \end{aligned}$$

That is, the optimal warehouse location is a convex combination of the distribution center locations.

- (e) Suppose $\lambda_i > 0$ for some index i . What does this imply about the corresponding point p_i in relation to the minimal enclosing circle?